

Convergent Inner-Outer Approximation Schemes from de Finetti Theorems for Games and Quantum Error Correction

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Polynomial Optimization

The setting: polynomial optimization

Many quantum and GPT problems take the form

$$\begin{aligned} p^* = \inf_{(x_A, x_B) \in K_A \times K_B} & \quad p(x_A, x_B) \\ \text{s. t.} & \quad f(x_A, x_B) = 0, \\ & \quad g(x_A, x_B) \geq 0, \end{aligned} \tag{1}$$

where K_A, K_B are **convex bodies** (state spaces) and p, f, g are affine.

- ▶ generally intractable [Gha10; Ioa07]
- ▶ includes quantum separability, non-local game values, and many GPT problems [DPS04; Ber+21; PLG25; Tav+24]

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Goal: an outer and inner approximation scheme with **quantitative** convergence guarantees, compatible with **equality and inequality** constraints and **numerically** implementable

From affine to linear: tensor product reformulation

For affine $f : K_A \times K_B \rightarrow V$ there is a unique linear $\hat{F}_{\hookrightarrow} : K_A \dot{\otimes} K_B \rightarrow V$.

We obtain the equivalent problem

$$\begin{aligned} p^* = \inf_{x_A \otimes x_B \in K_A \dot{\otimes} K_B} & \quad \hat{P}_{\hookrightarrow}(x_A \otimes x_B) \\ \text{s. t.} & \quad \hat{F}_{\hookrightarrow}(x_A \otimes x_B) = 0, \\ & \quad \hat{G}_{\hookrightarrow}(x_A \otimes x_B) \geq 0. \end{aligned} \tag{2}$$

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Two natural tensor products [Aub+21; Plá23]:

- ▶ **minimal:** $K_A \hat{\otimes} K_B = \text{conv}\{x_A \otimes x_B : x_A \in K_A, x_B \in K_B\}$
(separable)
- ▶ **maximal:** $K_A \hat{\otimes} K_B$ (all entangled states)

Four flavours of the optimization problem

p_1 : products in $K_A \dot{\otimes} K_B$

$$\begin{aligned} & \inf \widehat{P}_{\hookrightarrow}(x_A \otimes x_B) \\ & \text{s.t. } x_A \otimes x_B \in K_A \dot{\otimes} K_B \\ & \widehat{F}_{\hookrightarrow}(x_A \otimes x_B) = 0, \widehat{G}_{\hookrightarrow}(x_A \otimes x_B) \geq 0 \end{aligned}$$

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p_4 : all of $K_A \hat{\otimes} K_B$

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$$p_4 \leq p_3 \leq p_1 = p_2$$

The gap $p_1 \neq p_3$ is the subtle point: constraints need not pass to extreme points.

DPS-type hierarchy: symmetric extensions

A state $x_{AB} \in K_A \hat{\otimes} K_B$ is **n -extendible** if it has a permutation-invariant extension on $K_A \hat{\otimes} K_B^{\hat{\otimes} n}$ [DPS04; BL09].

- ▶ Denote the set by $\text{Ext}_n(K_A, K_B)$.
- ▶ $\text{Ext}_{n+1} \subseteq \text{Ext}_n$, and by [AMP24]

$$\bigcap_{n \geq 1} \text{Ext}_n(K_A, K_B) = K_A \dot{\otimes} K_B. \quad (3)$$

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$\Rightarrow$ DPS hierarchy converges to p_3 **asymptotically**.

But: convergence holds only **on average**.

- ▶ $\widehat{F}_{\hookrightarrow}(x_{AB}) = 0$ does **not** imply $\widehat{F}_{\hookrightarrow}(x_A^\lambda \otimes x_B^\lambda) = 0$ for every λ .
- ▶ No finite-level error bounds.
- ▶ Inequality constraints are not handled.

The polarization hierarchy [PLG25]

Idea: symmetric **double** extensions $(K_A \hat{\otimes} K_B)^{\hat{\otimes} n}$, plus a **polarization** constraint enforcing equality on extreme points [PLG25; LG23].

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- ▶ No **inequality** constraints.
- ▶ Heuristic interior points only.

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This work: a third route via information theory.

Comparison at a glance

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Constraints	global equality	local eq. & ineq.
Conv. rate	asymptotic	quantitative finite- n
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We restrict the constraint structure to the **local** form

$$f(x_A, x_B) = f_A(x_A) \oplus f_B(x_B), \quad (4)$$

i.e. $F_A(x_A) = 0$, $F_B(x_B) = 0$, $G_A(x_A) \geq 0$, $G_B(x_B) \geq 0$.

A convergent hierarchy for polynomial optimization

From the finite de Finetti theorem to a hierarchy

Recap. For any permutation-symmetric $x_{AB_1^n} \in K_A \hat{\otimes} K_B^{\hat{\otimes} n}$, there is a separable $\tilde{x}_{AB} \in K_A \dot{\otimes} K_B$ with

$$\|\tilde{x}_{AB} - x_{AB}\| \leq \frac{c(A, B)}{\sqrt{n}}, \quad (5)$$

where $c(A, B)$ depends only on K_A, K_B [Zei+26].

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Strategy. Relax the intractable problem over $K_A \dot{\otimes} K_B$ to one over the **symmetric extensions**:

- ▶ enforce the local constraints **on one B -extension**;
- ▶ use the de Finetti bound to translate this into an explicit error in $\sqrt{n}^{-1}$.

The hierarchy

For $n \in \mathbb{N}$, define the level- n outer relaxation

$$p^{(n)} = \inf_{x_{AB_1^n}} \hat{P}_{\hookrightarrow}(x_{AB_1})$$

$$\text{s.t. } x_{AB_1^n} \in K_A \hat{\otimes} K_B^{\hat{\otimes} n},$$

$$x_{AB_1^n} \text{ permutation-invariant on } B_1, \dots, B_n,$$

$$\hat{F}_{A, \hookrightarrow} \otimes \text{id}_{B^n}(x_{AB_1^n}) = 0, \quad \text{id}_A \otimes \hat{F}_{B, \hookrightarrow} \otimes \text{id}_{B^{n-1}}(x_{AB_1^n}) = 0,$$

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 \end{aligned}$$

Outer hierarchy:

$$p^{(1)} \leq p^{(2)} \leq \dots \leq p^{(n)} \leq \dots \leq p_3. \quad (6)$$

Key difference to DPS [DPS04]: the local constraints are enforced **on every marginal**, not only on average.

Convergence with explicit finite- n rate

Theorem (Convergence of the hierarchy [Koß+25; Zei+25])

Under mild regularity assumptions on $K_A, K_B, p, f_A, f_B, g_A, g_B$,

$$0 \leq p_3 - p^{(n)} \leq \frac{2 \left\| \hat{P}_{\hookrightarrow} \right\|_{1_{K_A \hat{\otimes} K_B}} \cdot c(A, B)}{\sqrt{n}} \quad (7)$$

where $c(A, B)$ is the de Finetti constant depending on K_A, K_B .

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where $c(A, B)$ is the de Finetti constant depending on K_A, K_B .

- ▶ First **quantitative finite-level guarantee** for general convex bodies.
- ▶ Handles **equality and inequality** constraints simultaneously.
- ▶ Recovers DPS/polarization rates in the quantum case [BH17; Chr+07].

Generating inner points

The level- n outer solution $x_{AB_1^n}^{(n)}$ is not yet a feasible product state.

Rounding scheme (cf. [Koß+25; Zei+25]):

- 1 Take the de Finetti candidate $\tilde{x}_{AB} = \sum_{\lambda} p_{\lambda} x_A^{\lambda} \otimes x_B^{\lambda}$ obtained from $x_{AB_1^n}^{(n)}$ by **measurement rounding**
- 2 Among the components $\{x_A^{\lambda} \otimes x_B^{\lambda}\}_{\lambda}$, pick the one minimizing the objective subject to the local constraints.

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Output: a certified **feasible** interior point $x_A^{\star} \otimes x_B^{\star} \in K_A \dot{\otimes} K_B$.

Conceptually: the de Finetti theorem provides a **constructive bridge** from the outer relaxation to a genuine feasible (primal) point.

Sandwich bound on the original value

Theorem (Convergent inner sequence [Koß+25; Zei+25])

Let $p_{\text{in}}^{(n)}$ be the objective value at the rounded interior point. Then

$$p^{(n)} \leq p_3 \leq p_{\text{in}}^{(n)}, \quad p_{\text{in}}^{(n)} - p^{(n)} \leq \mathcal{O}\left(\frac{1}{\sqrt{n}}\right). \quad (8)$$

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Consequence – a true sandwich:

$$p^{(n)} \text{ (outer, SDP/conic)} \longrightarrow \leq p_3 \leq \longleftarrow p_{\text{in}}^{(n)} \text{ (inner, certified)}$$

Both bounds tighten at the rate $1/\sqrt{n}$. We obtain matching upper and lower bounds on the true optimum.

Representability: when can we compute the hierarchy?

The hierarchy is a **convex** optimization problem over $K_A \hat{\otimes} K_B^{\hat{\otimes} n}$.

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Examples:

- ▶ Quantum state spaces $\rightarrow$ SDP [DPS04; Wil16].
- ▶ Classical simplices $\rightarrow$ LP.
- ▶ Homogeneous self-dual cones $\rightarrow$ symmetric cone programming [GIL25; HN08; GPT13; Tho19].

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$\Rightarrow$ for these state spaces the entire scheme is **numerically implementable**.

Application: non-local games

Two-player non-local games

A non-local game is specified by:

- ▶ input distribution $\pi(x, y)$ on questions $(x, y) \in \mathcal{X} \times \mathcal{Y}$,
- ▶ win predicate $V(a, b \mid x, y) \in \{0, 1\}$ for answers $(a, b) \in \mathcal{A} \times \mathcal{B}$.

GPT value of the game over state spaces K_A, K_B :

$$\omega_{\text{GPT}} = \sup_{x_{AB}, \{e_a^x\}, \{f_b^y\}} \sum_{x, y, a, b} \pi(x, y) V(a, b \mid x, y) \langle e_a^x \otimes f_b^y, x_{AB} \rangle, \quad (9)$$

where the sup is over $x_{AB} \in K_A \dot{\otimes} K_B$ and **local measurements** $\{e_a^x\} \subseteq E(K_A)$, $\{f_b^y\} \subseteq E(K_B)$.

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- ▶ $K_A, K_B = \text{classical} \Rightarrow$ Bell local value.
- ▶ $K_A, K_B = \text{quantum} \Rightarrow$ quantum value [Tav+24; Eke91].
- ▶ General GPTs: a wide spectrum in between [Plá23; BL09].

Non-local games as polynomial optimization

Reformulation. Treating measurements as variables in dual cones, the GPT value can be written as

$$\begin{aligned} \omega_{\text{GPT}} = \sup \quad & p(x_A, x_B) \\ \text{s.t.} \quad & x_A \in K'_A, \ x_B \in K'_B, \\ & f_A(x_A) = 0, \ f_B(x_B) = 0, \\ & g_A(x_A) \geq 0, \ g_B(x_B) \geq 0, \end{aligned} \tag{10}$$

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Crucial fact: the constraints are **local** – they encode the simplex structure of measurements on each side separately.

$\Rightarrow$ falls precisely into our framework.

Approximation scheme for non-local games

Applying the approximation hierarchy yields:

Theorem (GPT-value approximation)

For any two-player non-local game over state spaces K_A, K_B admitting conic lifts, there is a sequence of conic programs producing

$$\omega_{\text{GPT}}^{(n)} \rightarrow \omega_{\text{GPT}} \quad \text{with rate } \mathcal{O}(1/\sqrt{n}), \quad (11)$$

*together with certified **feasible** strategies achieving values $\omega_{\text{GPT}} - \mathcal{O}(1/\sqrt{n})$.*

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In particular (cf. [Jee+21; Zei+25; Ber+21]):

- ▶ For quantum systems of bounded dimension: **symmetry-based** approximation algorithms with **$\text{poly}(\epsilon^{-1})$** performance.
- ▶ For weakly self-dual GPTs: explicit rates in terms of geometric data of K_A, K_B .
- ▶ Inequality constraints (e.g. no-signalling, dimension) are handled **natively**.

Conclusion

What we did.

- ▶ Built an information-theoretic finite de Finetti theorem on convex bodies.
- ▶ Used it to obtain a convergent outer hierarchy with **explicit $1/\sqrt{n}$ rate**.
- ▶ Constructed certified **inner points** by measurement rounding to the de Finetti candidate.
- ▶ Sandwiched the optimum of polynomial optimization problems and recast non-local games inside the framework.

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Open directions.

- ▶ More general global constraints [PLG25].
- ▶ Other applications (see e.g. approximate quantum error correction [Koß+25] and channel discrimination [Ohs+24]).

Thank you!

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